Chapter 5: Trees

Technical / interview questions, exercises, multiple choice questions and answer key for Chapter 5 (Trees) of Data Structures.

Technical / Interview Questions

1. What are Non-Linear Data Structures? How do they differ from linear structures?
2. List the characteristics of Non-Linear Data Structures.
3. Give real-life examples where non-linear structures are used.
4. Why are trees considered hierarchical data structures?
5. Define a tree. What are root, leaf, parent, child, and sibling?
6. What is the degree of a node and degree of a tree?
7. Define height and depth of a tree.
8. What is a subtree?
9. Differentiate between full, complete, and perfect binary trees.
10. What is a skewed tree?
11. What is a balanced binary tree?
12. What is the difference between Binary Tree and Binary Search Tree?
13. Explain array representation of binary trees.
14. Explain linked representation of binary trees.
15. Compare array vs linked representation.
16. What are common operations performed on trees?
17. Explain tree traversal and its types.
18. What is the time complexity of tree traversal?
19. Define Binary Tree and its properties.
20. What is the maximum number of nodes in a binary tree of height h?
21. Explain Left-Child Right-Sibling (LCRS) representation.
22. What are the steps to convert a general tree to binary tree?
23. Differentiate between Inorder, Preorder and Postorder.
24. Where is each traversal used in real applications?
25. Why is BST preferred for searching operations? Explain time complexity in best and worst case.

Programming Exercises

Write programs in C / Java / Python:

1. Implement a basic structure for a tree node.
2. Write a function to calculate the height of a binary tree.
3. Implement a function to create a mirror image of a binary tree.
4. Convert a binary tree to a doubly linked list in-place.
5. Write a function to check if a binary tree is balanced.
6. Implement a function to find the lowest common ancestor of two nodes in a binary tree.
7. Write functions for pre-order, in-order, and post-order traversal of a binary tree.
8. Find the maximum sum from any node to any node in a binary tree.
9. Calculate the diameter of a binary tree.

10. Write a function to validate if a binary tree is a Binary Search Tree.
11. Find the kth smallest element in a Binary Search Tree.
12. Operations on Binary Search Trees:
13. Implement the insertion operation for a Binary Search Tree.
14. Implement the deletion operation for a Binary Search Tree.
15. Implement iterative versions of pre-order, in-order, and post-order tree traversal.
16. Create a binary tree.
17. Count total nodes in a tree.
18. Count leaf nodes.
19. Calculate height of tree.
20. Perform inorder traversal.
21. Perform preorder traversal.
22. Perform postorder traversal.
23. Level order traversal using queue.
24. Find maximum element in tree.
25. Mirror a binary tree.
26. Create BST.
27. Insert element into BST.
28. Search element in BST.
29. Delete a node from BST.
30. Find minimum value in BST.
31. Find maximum value in BST.
32. Count nodes in BST.
33. Validate whether given tree is BST.
34. Find kth smallest element in BST.
35. Find Lowest Common Ancestor in BST.
36. Convert general tree to binary tree (LCRS).
37. Check if binary tree is balanced.
38. Find diameter of tree.
39. Find depth of a node.
40. Print all root-to-leaf paths.

Multiple Choice Questions

1. What is a tree in data structures?

(a) A collection of nodes connected hierarchically
(b) A linear data structure
(c) A loop in a program
(d) A type of sorting algorithm

2. Which of the following is a non-linear data structure?

(a) Array
(b) Linked List
(c) Tree
(d) Stack

3. The topmost node of a tree is called:

(a) Leaf
(b) Parent
(c) Root
(d) Child

4. Nodes with no children are called:

(a) Internal nodes
(b) Root nodes
(c) Leaf nodes
(d) Sibling nodes

5. The maximum number of children a node in a binary tree can have is:

(a) 1
(b) 2
(c) 3
(d) 4

6. Inorder traversal of a binary tree follows which order?

(a) Root – Left – Right
(b) Left – Root – Right
(c) Left – Right – Root
(d) Right – Root – Left

7. Preorder traversal follows which order?

(a) Root – Left – Right
(b) Left – Root – Right
(c) Left – Right – Root
(d) Right – Left – Root

8. Postorder traversal follows which order?

(a) Root – Left – Right
(b) Left – Root – Right
(c) Left – Right – Root
(d) Root – Right – Left

9. In a Binary Search Tree (BST), the left child of a node contains:

(a) Values greater than the root
(b) Values less than the root
(c) Any random value
(d) Duplicate values only

10. The right child of a BST node contains:

(a) Values less than the root
(b) Equal values only
(c) Values greater than the root
(d) No values

11. Searching in a balanced BST has time complexity:

(a) O(1)
(b) O(n)
(c) O(log n)
(d) $O(n^2)$

12. Worst-case time complexity of searching in a BST is:

(a) O(1)
(b) O(log n)
(c) O(n)
(d) $O(n^2)$

13. A complete binary tree is one in which:

(a) All nodes have two children
(b) All levels are filled except possibly the last
(c) Only leaf nodes exist
(d) It has no root

14. A perfect binary tree is one in which:

(a) All internal nodes have one child
(b) All leaf nodes are at the same level
(c) Only one node exists
(d) Nodes are randomly arranged

15. The degree of a leaf node is:

(a) 0
(b) 1
(c) 2
(d) 3

16. Level order traversal uses which data structure?

(a) Stack
(b) Queue
(c) Array
(d) Recursion only

17. The height of a tree with only one node is:

(a) 0
(b) 1
(c) 2
(d) Undefined

18. In array representation of a binary tree, it is most efficient for:

(a) Skewed trees
(b) Complete binary trees
(c) Sparse trees
(d) Cyclic graphs

19. LCRS stands for:

(a) Left Child Right Sibling
(b) Level Child Root Structure
(c) Linked Child Root System
(d) Logical Child Recursive Structure

20. Deleting a node with two children in BST requires:

(a) Removing directly
(b) Replacing with inorder successor or predecessor
(c) Swapping with root
(d) Ignoring children

21. A skewed binary tree resembles:

(a) A complete tree
(b) A linked list
(c) A perfect tree
(d) A balanced tree

22. The maximum number of nodes at level i in a binary tree is:

(a) i
(b) 2i
(c) 2^i
(d) i^2

23. The maximum number of nodes in a binary tree of height h is:

(a) h^2
(b) 2h
(c) $2^{h+1} - 1$
(d) h + 1

24. A subtree is:

(a) A separate graph
(b) A part of a tree consisting of a node and its descendants
(c) A cycle
(d) A linear structure

25. Inorder traversal of a BST gives:

(a) Reverse order
(b) Sorted order
(c) Random order
(d) Level order

26. The root node has:

(a) No parent
(b) No children
(c) Two parents
(d) Cycles

27. Internal nodes are nodes that:

(a) Have no children
(b) Have at least one child
(c) Are always root
(d) Have two parents

28. A full binary tree is one in which:

(a) Every node has 0 or 2 children
(b) Every node has 1 child
(c) Every node has 3 children
(d) Only leaf nodes exist

29. The average-case time complexity of BST insertion is:

(a) O(1)
(b) O(log n)
(c) $O(n^2)$
(d) $O(n^3)$

30. Trees are useful for representing:

(a) Sequential data only
(b) Hierarchical relationships
(c) Simple loops
(d) Mathematical equations only

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	2	C	3	C	4	C	5	B
6	B	7	A	8	C	9	B	10	C
11	C	12	C	13	B	14	B	15	A
16	B	17	A	18	B	19	A	20	B
21	B	22	C	23	C	24	B	25	B
26	A	27	B	28	A	29	B	30	B